

(a) TRAINING

FIRE train set

349 official pairs

GitHub PLC

permissive open source

Filter & convert

licence filter · AST parse
DeepSeek-V3 NL · augment

744 NL-AST pairs

396 pass quality checks
356 train / 40 held-out

LoRA fine-tuning

Qwen2.5-Coder 1.5B / 7B
 $r=16$, $\alpha=32$

Trained adapter

NL $\rightarrow$ compact AST

(b) INFERENCE

NL task spec

text + I/O address list

Fine-tuned LLM

greedy decoding

Compact JSON AST

```
{ "t": "Asn", "tg": "Pump",  
  "v": { "t": "Id", ... } }
```

Parse layer

strip fences · json_repair
 $\rightarrow$ typed AST

Static linter

8 rules: missing/unused
I/O, input writes, ...

Scan simulator

200 scan cycles; flags
outputs that never change

AST verification

Clean?

issues or clean

same verified AST

yes

Task A

IEC 61131-3 ST

ST printer

deterministic

Rust printer

Modbus scan loop

Rust post-pass

Modbus stub + rustc

Task B

Rust program

 learned

 deterministic code

 verification

 final output

 self-repair feedback